


SAPIENZA
UNIVERSITÀ DI ROMA

# A Hybrid Mixture-of-Experts Architecture for Sensor-Attack Detection in Water Distribution Networks

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MSc in Data Science

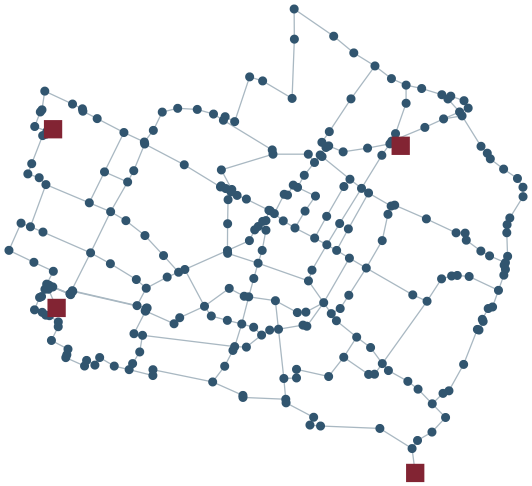
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# Water distribution networks (WDNs)

Pipes connect water sources to consumers.
Pressure sensors monitor conditions in the network.

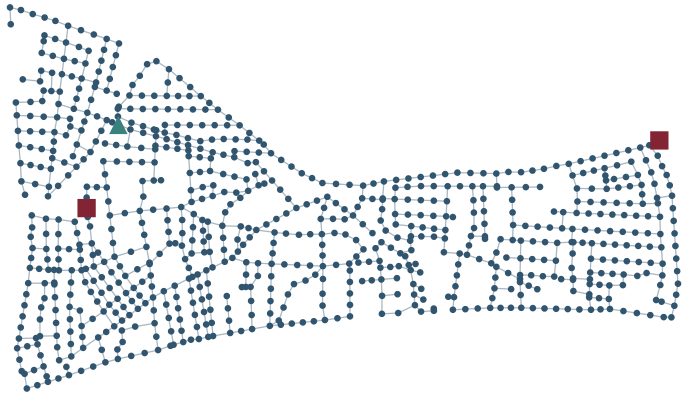
## Modena



### 272 nodes, 317 links

268 junctions, 4 reservoirs
317 pipes

## L-Town



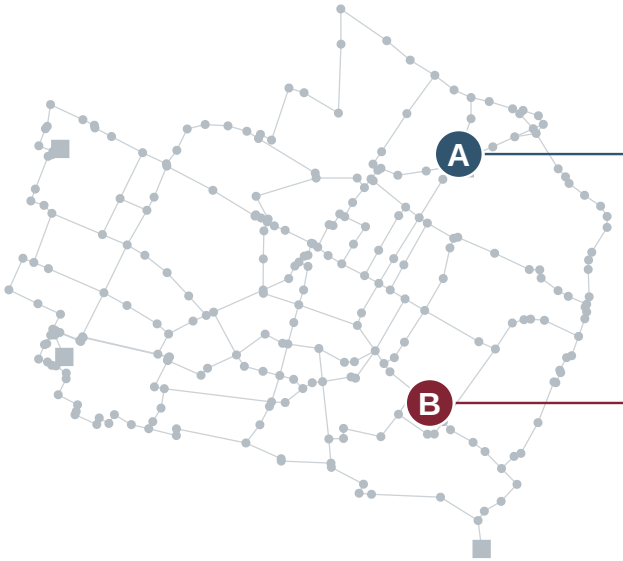
### 785 nodes, 909 links

782 junctions, 2 reservoirs, 1 tank
905 pipes, 1 pump, 3 valves

# Problem statement

## Detect five types of sensor-data attacks in water distribution networks.

Modena: normal and attacked sensor examples




### A Normal sensor

Reports an unaltered pressure reading.

### B Attacked sensor

Reports a pressure reading modified by an attack.

**Random, targeted, replay, drift and noise**

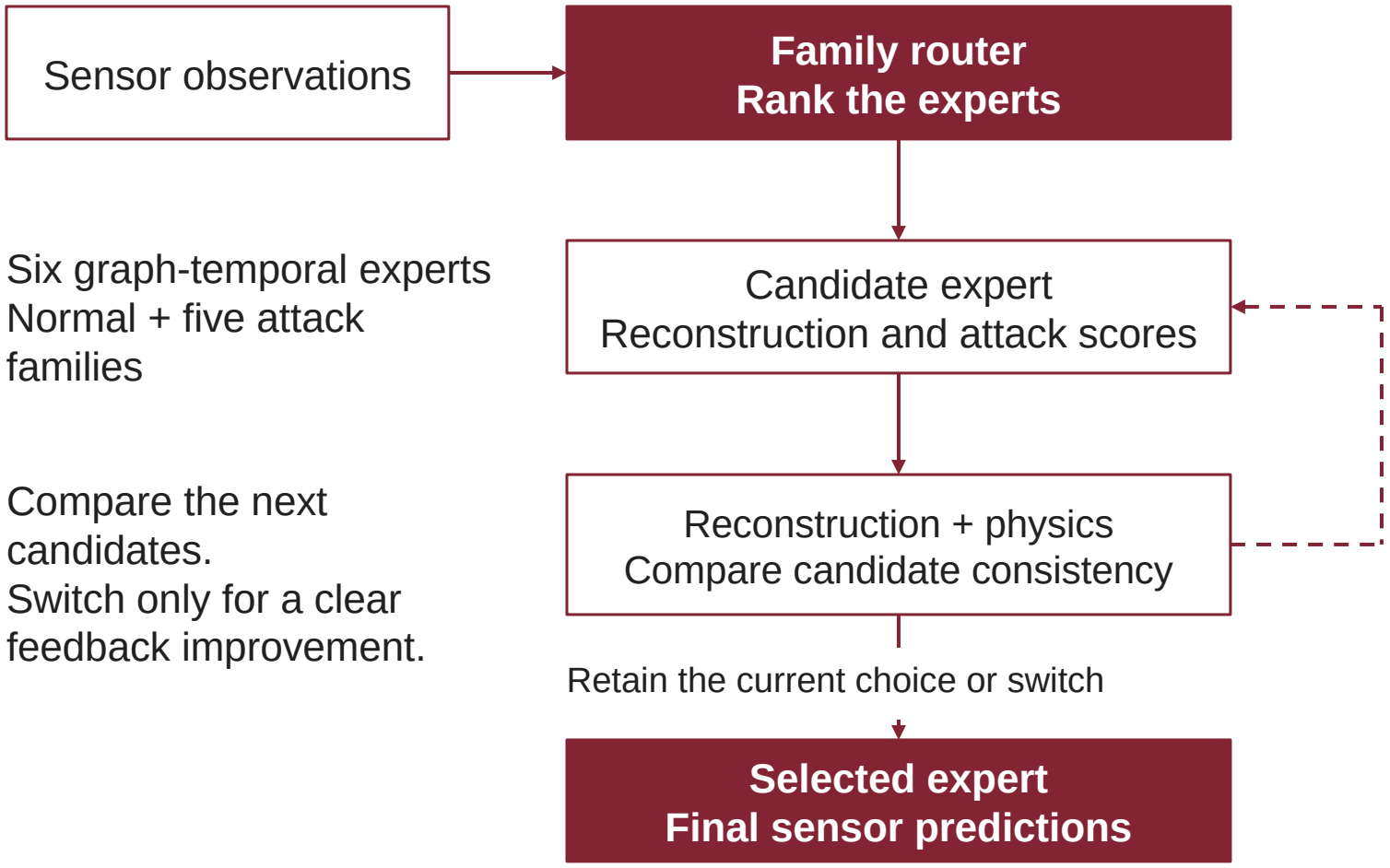
# Five attack mechanisms

We modify the pressure readings received by the detector.

| Family | How the reading changes | Experimental setting |
|---|---|---|
| Random | Persistent bias at randomly selected sensors | Magnitude: 0.5–2.0 m |
| Targeted | Persistent bias at sensors selected by node degree | Magnitude: 0.5–2.0 m |
| Replay | An earlier observation replaces the current reading | Delay: 2–6 h |
| Drift | A bias increases gradually | Target: 0.5–2.0 m<br>Ramp: 6–18 h |
| Noise | Random perturbations increase variability | Extra noise SD:<br>0.3–0.8 m |

Pressure measurement noise: 0.1 m standard deviation (SD).

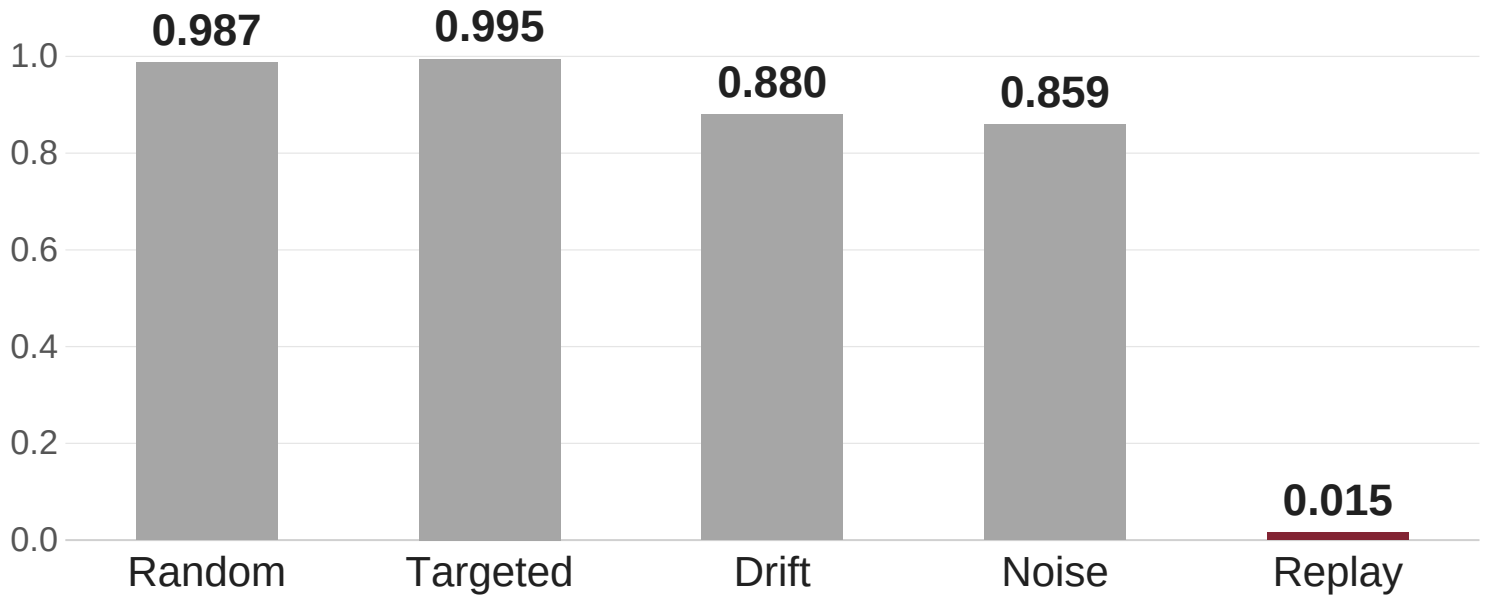
# The earlier cascade architecture



# Why the architecture changed

## Early cascade: attack-family F1

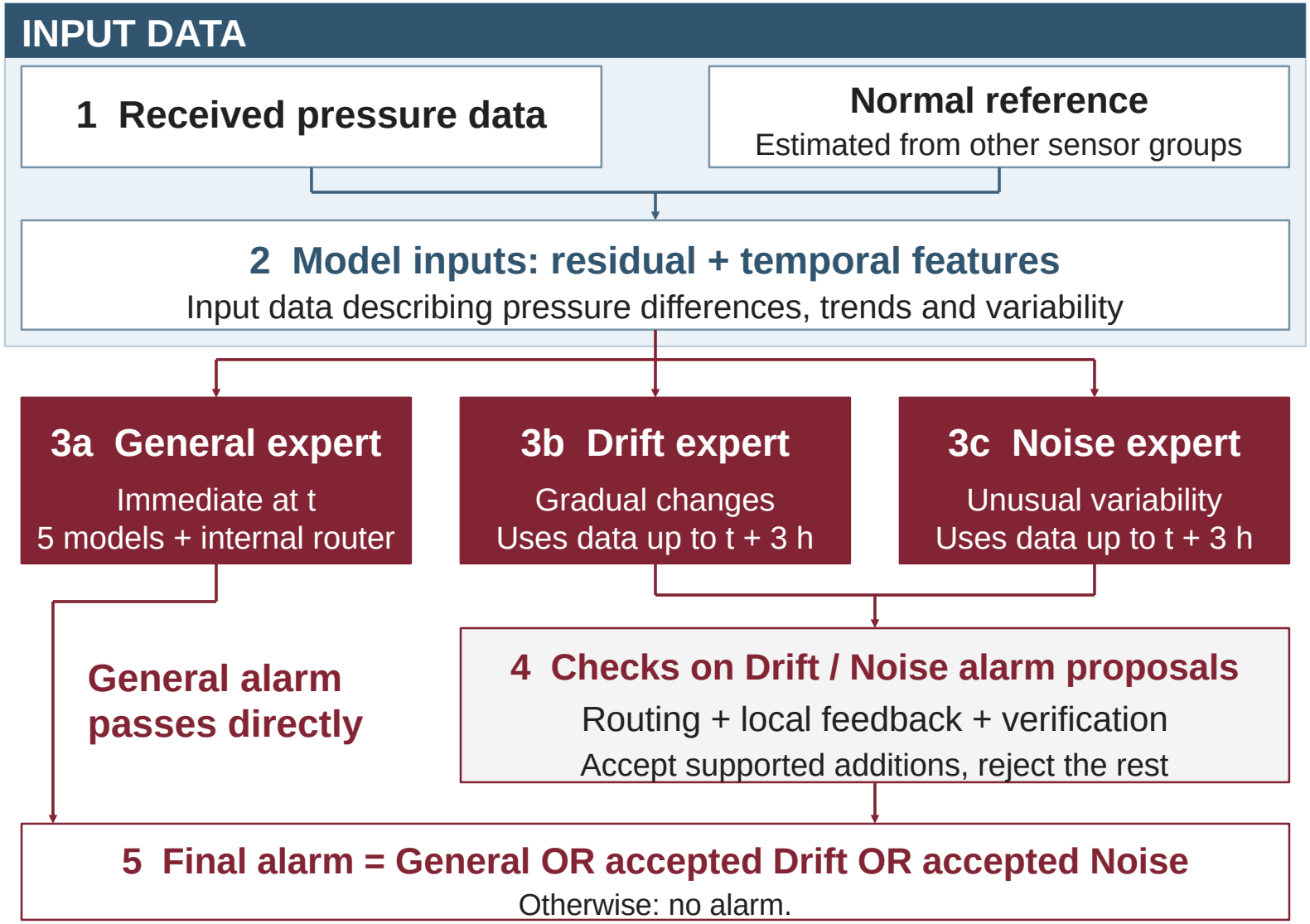
Modena development experiment, mean across 10 fitted runs




**Four families performed well. Replay F1 was only 0.015.**

This motivated the search for reliable coverage across all five families.

# Full hybrid workflow



# How we tested the detector

## 1 Simulate normal operation

Generate pressure readings for Modena and L-Town.

## 2 Modify selected sensor readings

Apply five attack types and record which readings change.

## 3 Run the trained detector

Use separate evaluation scenarios and fixed thresholds.

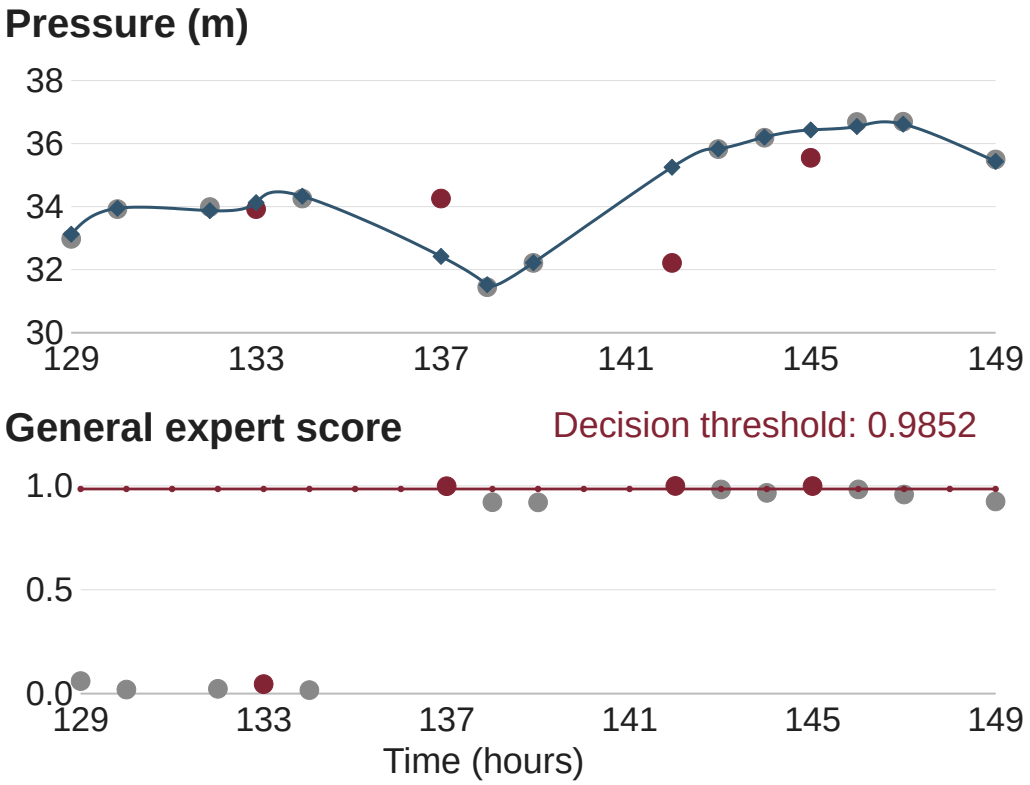
## 4 Compare alarms with known labels

Count detections, missed attacks and false alarms.

# Replay detection in one sensor

Modena, sensor 18: a reading from 3 hours earlier

**Normal reference** **Unmodified reading** **Replayed reading**




## At hour 137

Received: 34.26 m
Expected: 32.42 m

The expert evaluates this difference and its recent history.

**Score: 0.9986**
**Threshold: 0.9852**
**Alarm raised**

**Hybrid: 3 of 4 altered readings detected. 1 missed. 1 false alarm.**

# Final results on two networks

## Final evaluation: 10 model fits × 6 sources per network

| Mean metric | Modena | L-Town |
|---|---|---|
| Pooled F1 | 0.8669 | 0.8950 |
| Family macro F1 | 0.9045 | 0.9083 |
| Clean FPR (%) | 0.0584 | 0.0550 |

**Modena: all five family means exceed 0.80.**
**L-Town: four exceed 0.89; replay remains at 0.7312.**

168-hour scenarios; hourly readings; 50% pressure missingness.
24 scenarios per source. The 60 model–source cells are not independent.

# Later refinements in each network

## Modena: original confirmation with 5 model fits

| Pooled F1 | Earlier hybrid | Threshold only | Integrated |
|---|---|---|---|
| 5 fits × 6 sources | 0.6863 | 0.6998 | 0.8672 |

False positives fall by 84.2%, with 4.1% fewer true positives.
Drift and noise F1 fall slightly.
Extending to 10 fits gives the final mean of 0.8669.

## L-Town: history comparison with 10 model fits

| Mean metric | No history | History |
|---|---|---|
| Replay F1 | 0.7065 | 0.7312 |
| Clean FPR (%) | 0.0627 | 0.0550 |

Replay improves in all 60 pairs, but still misses the 0.75 goal.

# Overall scores against LightGBM

Same evaluation sources and three-hour information allowance

| Network | Metric | LightGBM | Hybrid |
|---|---|---|---|
| Modena | Family macro F1 | 0.8593 | **0.9045** |
| | Pooled F1 | 0.8509 | **0.8669** |
| | Clean FPR (%) | **0.0317** | 0.0584 |
| L-Town | Family macro F1 | 0.8823 | **0.9083** |
| | Pooled F1 | **0.8977** | 0.8950 |
| | Clean FPR (%) | **0.0184** | 0.0550 |

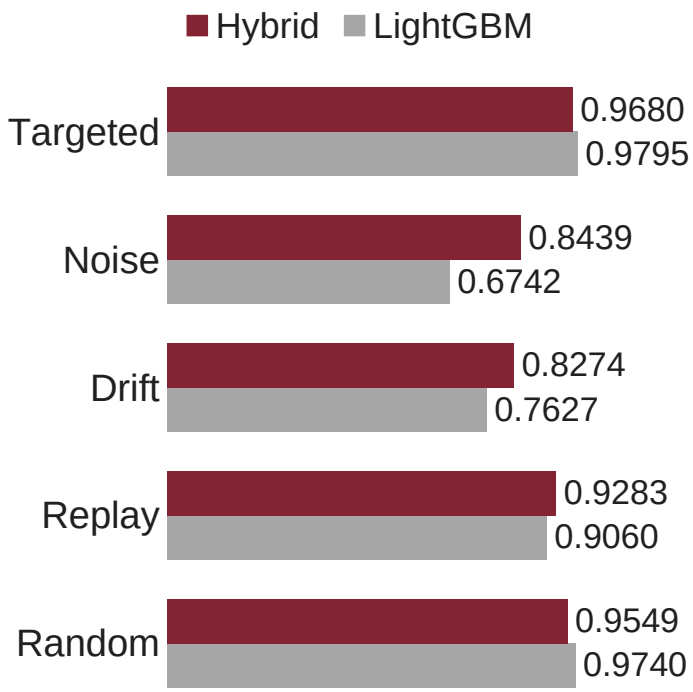
Original operating points. Bold marks the better score. Lower clean FPR is better.

Higher family macro F1 for the hybrid on both networks.
Fewer clean-period false alarms for the single LightGBM.

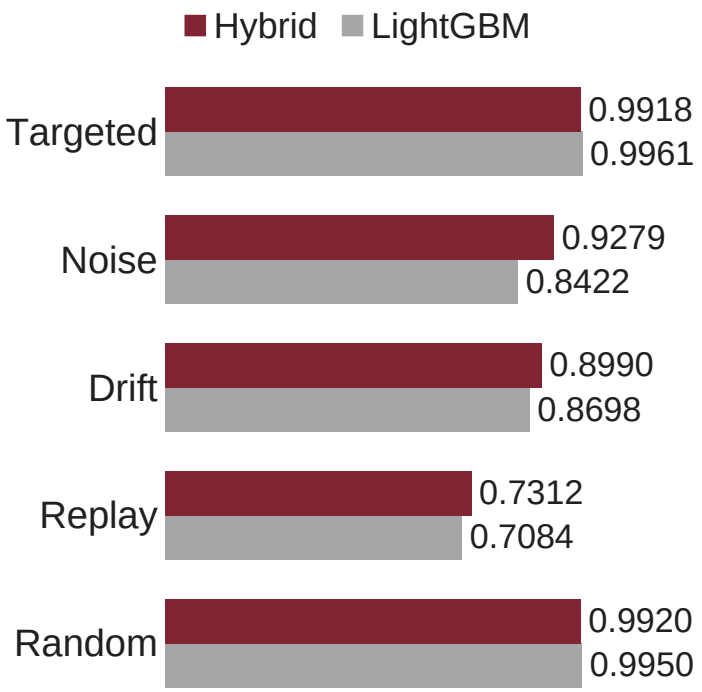
Pooled F1 favors the hybrid in Modena and slightly favors LightGBM in L-Town.

# Attack-family scores against LightGBM

## Modena




## L-Town




**Hybrid leads on replay, drift and noise in both networks.**

Mean family F1 at the reported operating points. LightGBM uses a 3-hour allowance.

# When is the hybrid useful?

## Stronger coverage of temporal attacks

At the reported operating points, replay, drift and noise F1 are higher on both networks than with single LightGBM.

## A measurable cost in false alarms

LightGBM has lower clean FPR in both networks.
It also leads on random and targeted attacks.

## The operating point matters

Calibration-derived threshold sweeps change the ranking.
The hybrid does not dominate across all operating points.

# Conclusions

## We built a detector for five sensor attack types.

Three experts analyse pressure data in parallel. General alarms pass directly. Additional Drift and Noise alarms require checks.

## We tested it on Modena and L-Town.

With simulated attacks and the reported thresholds, the hybrid had higher replay, drift and noise F1 than LightGBM on both networks. LightGBM raised fewer false alarms.

## The next step is testing with operational data.

Measure incident detection, alert delays and runtime before considering deployment.

# Thank you for your attention.

Any questions?